

Module 16: c-Bridge Certificate

Certified numerical observation: $c/10^6$ vs $\beta_0 = 299 + \pi/10$ | May 2026
Battle Plan v1.6 -- David Fox

This certificate records the precise numerical relationship between the speed of light c (SI definition, exact) and the transcendental constant $\beta_0 = 299 + \pi/10$ that defines the S_4 exceptional prime set. All values are computed at 100 decimal places using mpmath 1.3.0. No causal or physical claim is made -- this is a certified numerical observation.

Core Constants

Constant	Value	Source
$\beta_0 = 299 + \pi/10$	299.31415926535897932384626433832795...	mpmath $\pi / 10$ (100 dps); M1 SHA 63ef870a...
c (m/s)	299792458 [exact]	SI definition of the metre (1983)
$c / 10^6$	299.792458 [exact]	derived from c (m/s)

Ratio Analysis

Quantity	Certified Value (100 dps)
$r = (c/10^6) / \beta_0$	1.001597982320031113926463891128826083...
$\epsilon = r - 1$	0.001597982320031113926463891128826083...
$1 / \epsilon$	625.789151397200216...
$1/625$ [reference]	0.0016 [= 0.001599999... repeating]
$\epsilon - 1/625$	-0.000002017679968886074... [$\epsilon < 1/625$]
Relative gap $ \epsilon - 1/625 / (1/625)$	0.00126105... [0.13%]

Gap Analysis: The Interval [β_0 , 300]

The number 300 is the nearest integer above β_0 . Both $c/10^6$ and 300 are 'near-integer' perturbations of β_0 . The gaps are:

Gap	Value	Comment
$c/10^6 - \beta_0$	0.47829873464102067615...	How far $c/10^6$ overshoots β_0
$300 - \beta_0$	0.68584073464102067615...	How far 300 overshoots β_0
c fraction in $[\beta_0, 300]$	0.69739038596383... (69.74%)	c is ~70% of the way to 300
$\beta_0=300$ gives $C=193.4$	$g \leq 9348$ (M14)	Integer $\beta_0 \rightarrow$ trivial ($C \rightarrow \infty$)
$\beta_0=\beta_0$ gives $C=11.422$	$g \leq 33$ (M9/M10)	Transcendental $\beta_0 \rightarrow$ tight bound

The Nines Observation

The error $\epsilon = (c/10^6)/\beta_0 - 1 = 0.001597982...$ is close to $1/625 = 0.001600... = 0.001599999...$ (infinite repeating 9s). The proximity is within 0.13% (2.0e-6 absolute).

Expression	Value	Note
ϵ	0.0015979823200311...	$c/10^6 / \beta_0 - 1$
$1/625$	0.0016000000000000...	= 0.001599999... (repeating)
$1/\epsilon$	625.789151...	nearest integer: 626
$625 = 5^4$	base-10 terminating	$1/5^n$ always terminates in decimal
$ \epsilon - 1/625 $	2.018e-6	absolute gap
relative gap	0.126%	$(\epsilon - 1/625)/(1/625)$

Mathematical note: $1/625 = 0.0016$ exactly (terminating decimal). Every terminating decimal $d = 0.0016$ also equals $0.001599999...$ by the standard identity $x.y_1...y_n = x.y_1...(y_n - 1)999...$

Note on the RH Certificate Proposal

A separate conversation proposed using $\Delta_{DS}^{(4)} = 23.796910$ in an RH lower bound: $C(2) \geq (-10.756045 + 23.796910) / 1.511 = 6.629$. Module 15 (SHA cf1620c7...) has certified that $\Delta_{DS}^{(4)} = 2.753126$, not 23.796910 -- a factor-8.6 error. The RH pathway using the wrong Δ is NOT certified here. The c-Bridge observation stands independently.

Chain of Custody

Item	SHA-256
m16_c_bridge.py (script)	7940bda05307a37985033beabb1e51ede05770f3082c976bc276...
m16.out (stdout)	e1c042ba8df33a3b89046ca72c332c832f313eee2409b12963da...
Parent M1 (alpha_0)	63ef870a78766619327e99b68683bceff8c8ef9a525298756c77...
Parent M4 (S_4, p_5)	b810a7a331e47066e3eb4765a5ffdc17c1a56ddbff855a096c18...

Conclusion

Certified: $c/10^6 = 299.792458$ and $\beta_0 = 299 + \pi/10 = 299.31415...$ agree to within 1 part in 625.789. The ratio $r - 1 = 0.001597982...$ is within 0.13% of $1/625 = 0.001599999...$ (repeating nines). c is 69.74% of the way from β_0 to 300. Both are 'near-integer' perturbations of the transcendental β_0 . All values SHA-bound in m16.out. No physical interpretation is asserted.

CERTIFIED -- Battle Plan v1.6 -- David Fox -- May 2026